Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 37)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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6	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Describe coppicing and explain how it increases the biodiversity of a woodland.

[4 marks]

0 1 . 2 Explain how the reintroduction of grey wolves to Yellowstone National Park increased biodiversity.

[4 marks]

0 1 . 3 Explain why culling or removing species may be necessary in habitat management.

[2 marks]

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0 2 . 1 Explain the effects of deforestation on soil.

[4 marks]

0 2 . 2 Explain how deforestation affects hydrology and climate.

[4 marks]

0 2 . 3 Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

1

2

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0 3 . 1 Explain why introduced species often have a greater effect on island communities than on mainland communities.

[2 marks]

0 3 . 2 For each of the following, describe an example of an introduced species and explain its effect on an indigenous species.

[6 marks]

Introduced competitor

Introduced predator

Introduced pathogen

0 3 . 3 Explain how hybridisation with an introduced species threatens the red deer and the Scottish wildcat.

[2 marks]

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10

0 4 Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

Table 1

Feature	Site A: valley floor	Site B: south-facing slope
Frost-free period / days	110	150
Days with mean temperature above 5 °C	160	175
Mean summer rainfall / mm	180	150
Gradient / %	1	8

0 4 . 1 Use Table 1 to explain which site is more suitable for maize.

[3 marks]

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0 4 . 2 Describe **three** ways in which temperature can be controlled to increase crop productivity.

[3 marks]

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0 4 . 3 Explain why keeping livestock warm increases the amount of food converted to growth.

[2 marks]

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0 4 . 4 Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

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0 5 Table 1 compares extensive oyster farming with intensive salmon farming.

Table 1

Feature	Extensive oyster farm	Intensive salmon farm
Food input	None - filter natural plankton	Fishmeal / plant pellets
Stocking density	Low, spaced on seabed	High, in sea cages / tanks
Breeding	Wild spat settle on cleared seabed or hatchery	Selected adults stripped of eggs and milt
Disease / pest control	Removal of predatory crabs	Antibiotics, pesticides, wrasse, removal of diseased fish
Yield per hectare	Low	Very high

0 5 . 1 Use Table 1 to explain the relative merits of extensive and intensive aquaculture.

[4 marks]

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0 5 . 2 Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

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0 5 . 3 Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

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0 6 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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2 .....

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0 6 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

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0 6 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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5

0 7 A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

0 7 . 1 Calculate the estimated population size.

[2 marks]

Answer _____ beetles

0 7 . 2 State **four** assumptions of the Lincoln Index method.

[4 marks]

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0 7 . 3 Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

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0 7 . 4 Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

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0 8 . 1 Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

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Satellite / radio / GPS tracking .....

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DNA databases and eDNA .....

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Image recognition software .....

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Acoustic monitoring / sonograms .....

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0 8 . 2 Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

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0 9 Table 1 shows the numbers of breeding pairs of two seabird species on an island before and after a rat eradication programme.

Table 1

Species	Breeding pairs before eradication	Breeding pairs 8 years after eradication
Manx shearwater	45	310
Storm petrel	12	140

0 9 . 1 Explain why oceanic islands have many endemic species but low overall biodiversity.

[4 marks]

0 9 . 2 Use Table 1 to explain why introduced rats had such a severe effect on these birds.

[3 marks]

0 9 . 3 Explain how avian malaria threatens Hawaiian birds and how climate change worsens the threat.

[2 marks]

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0 9 . 4 Give **one** other conservation measure for oceanic islands besides eradicating introduced species.

[1 mark]

10

1 0 . 1 Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

Extra space

1 0 . 2 State what is meant by a **taxon**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

OR

1 1 . 2 Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS